QWeatherRiskF

Issue 2: Crop eligibility assessment for predictive modelling

Item	Detail
Input	crop_long_v1.csv (Issue 1 output, from merged_cleaned_v1f): 22 crops x 65 district-years = 1,430 rows, with quality flags
Outputs	crop_model_eligibility_v1.csv; crop_observation_coverage_summary_v1.csv; tables/ (observation, district and year coverage, sensitivity); this report
Reproduce	python crop_eligibility.py --input crop_long_v1.csv --outdir . (thresholds in CONFIG; saved in eligibility_config_used.json)

1. Summary

- **11 crops are eligible**: Groundnut, Maize, Rice, Sugarcane, Pigeonpea, Sesamum, Finger Millet, Chickpea, Pearl Millet, Sunflower, Rabi Sorghum. **3 are conditionally eligible**: Kharif Sorghum, Castor, Cotton. **5 data-poor crops are ineligible**: Soyabean, Rapeseed And Mustard, Linseed, Safflower, Wheat; the oilseeds and sorghum aggregates and the minor pulses group stay ineligible by the Issue 1 decision. Only the eligible crops proceed to predictive modelling (column proceed_to_modelling).
- **Five crops are eligible under every threshold setting tested** (rice, maize, groundnut, sugarcane, pigeonpea). The other six eligible crops become conditional only under the strictest sample-size rule (10 observations per predictor); the three conditional crops stay conditional or move one band under other settings (Figure 7).
- **Why the numbers are lower than the crop columns suggest:** a valid observation must be reported, positive, verifiable against production and area (rounding cannot move the implied yield by more than 10%) and not an allocated estimate. Zeros that were blanks in the source, tiny-area rows and cotton's eight estimated 2019 values do not count.
- **Several eligible crops are small:** sunflower, finger millet, pearl millet and sesamum together are about 2.6% of the area of the modelling crops. They pass the data criteria, but business relevance is for the team to confirm. Rice (44%), groundnut (16%) and cotton (12%) dominate.
- **Your notebook analysis is the basis for the missing / zero / valid breakdown** (Figure 1): its finding that blanks sat in 2016-17 and meant "not grown" is carried here as the category "Zero (was blank)", so these rows are never counted as valid.

2. Definitions and the valid-observation rule

Each district-year-crop row is assigned the first status that applies: **missing** (any of area, production, yield NaN); **zero** (all three zero, shown separately when the zero was a blank filled in v1d); **partly zero**; **unverifiable** (rounding of the stored area and production can change the implied yield production / area x 1000 by more than 10%); **yield inconsistent** (reported yield differs from production / area x 1000 by more than 5% and by more than the rounding bound); **estimated** (allocated, not observed); otherwise **valid**. Counts of each status per crop are in tables/crop_observation_coverage_summary.csv and row-level status in tables/observation_validity.csv.

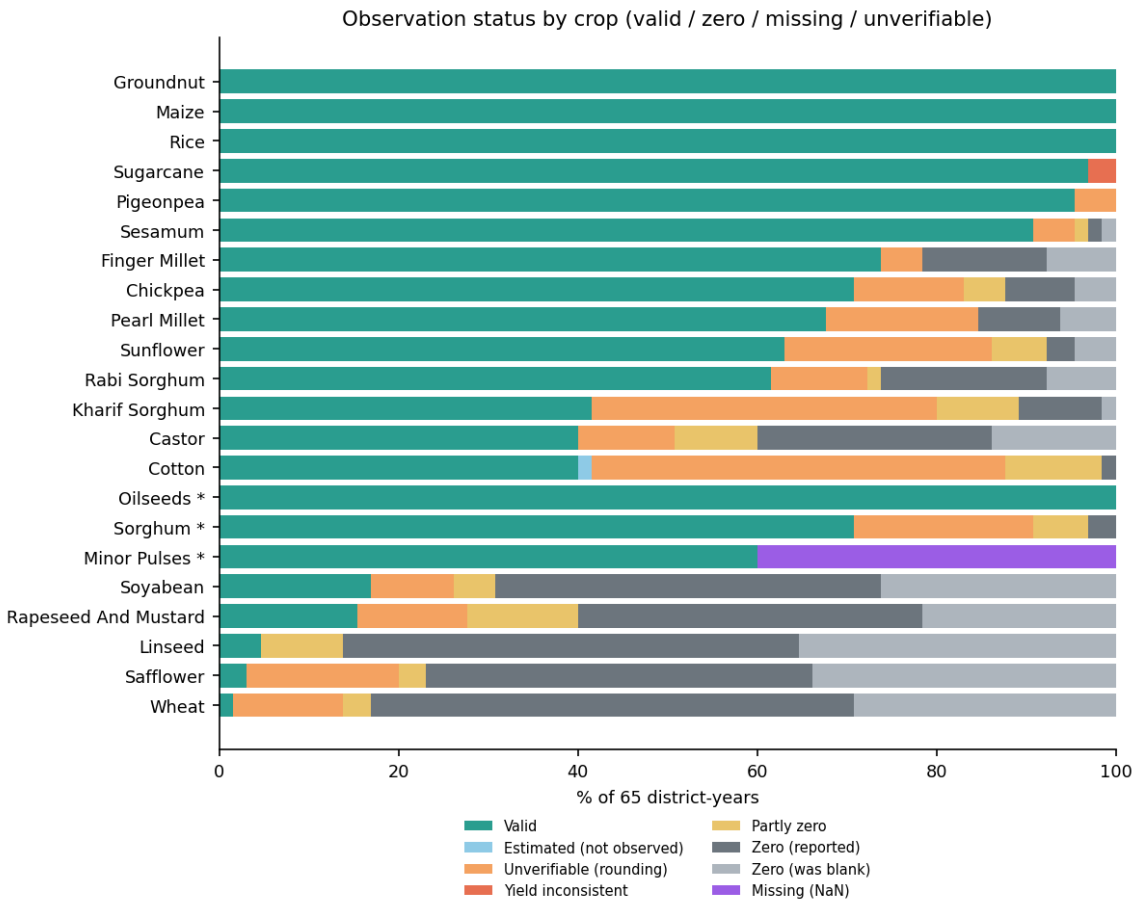


Figure 1. Observation status by crop. * = excluded by the Issue 1 decision (aggregate or group category). Compare with the missing / zero / valid chart in the cleaning notebook.

3. Yield consistency against production and area

Yield should equal production / area x 1000. The source stores area and production rounded (often to two decimals, cotton production to whole thousand tonnes), while yield is stored separately, so tiny areas produce apparent mismatches that are rounding, not error. Two checks were therefore used: a strict 5% test and a rounding-aware test that accepts a difference when it lies within the bound implied by the stored precision.

- **Strict test:** 50% of observations with area under 100 ha differ by more than 5%; the rate falls to 6% for 100-250 ha and 2% for 250-500 ha, and stays at about 3% or below above that (Figure 4, left).
- **Verifiability:** only 9% of observations below 100 ha can be checked to within 10%, against 98% for 250-500 ha and 93-100% above 500 ha (Figure 4, right). This is why a verifiability rule is used instead of an arbitrary minimum-area cut.
- **No extra area cut is imposed:** year-to-year yield instability (yield more than 1.5x or less than 0.67x the median of the same district and crop in other years) does not rise at small areas (share unstable: <=0.25 7%, 0.25-0.5 12%, 0.5-1 27%, 1-2 14%, 2-5 20%, 5-20 16%, >20 18%; tables/yield_instability_by_area_bin.csv), so there is no evidence to justify one. Rows below 1,000 ha are valid but flagged low-area in crop_long_v1.csv.
- **Result:** among verifiable positive observations, 100% of every crop except sugarcane (97%) pass the rounding-aware test; 2 sugarcane rows (Anantapuramu 2016, Prakasam 2018) remain inconsistent and are excluded. The strict rate is lower for sesamum (97%), finger millet (98%) and sunflower (98%), because of rounding at small areas.

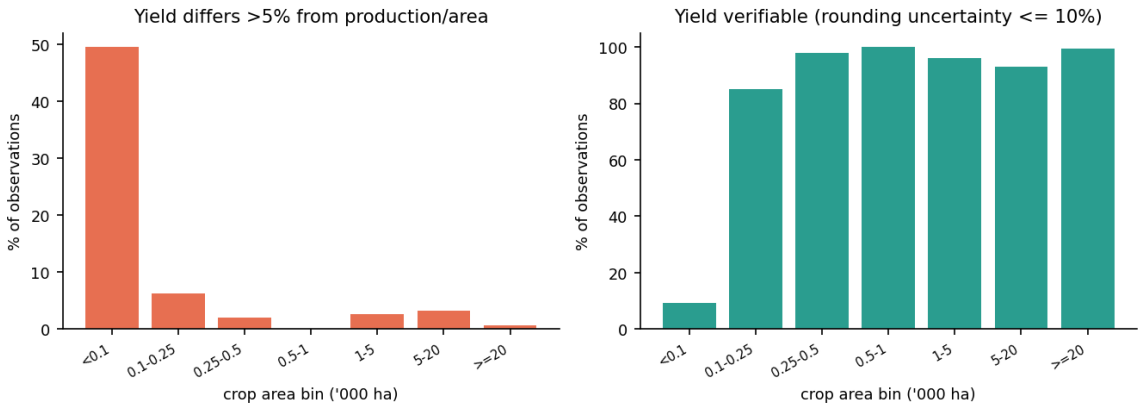


Figure 4. Left: share of observations where reported yield differs from production / area by more than 5%. Right: share of observations whose implied yield is verifiable to within 10%.

4. Coverage by district and year

A district is **usable** for a crop if it has at least 4 valid years, so that after holding one year out a district reference mean still has 3 training years. Year coverage is the smallest number of valid districts in any single year, which sets the size of a held-out test year in leave-one-year-out validation.

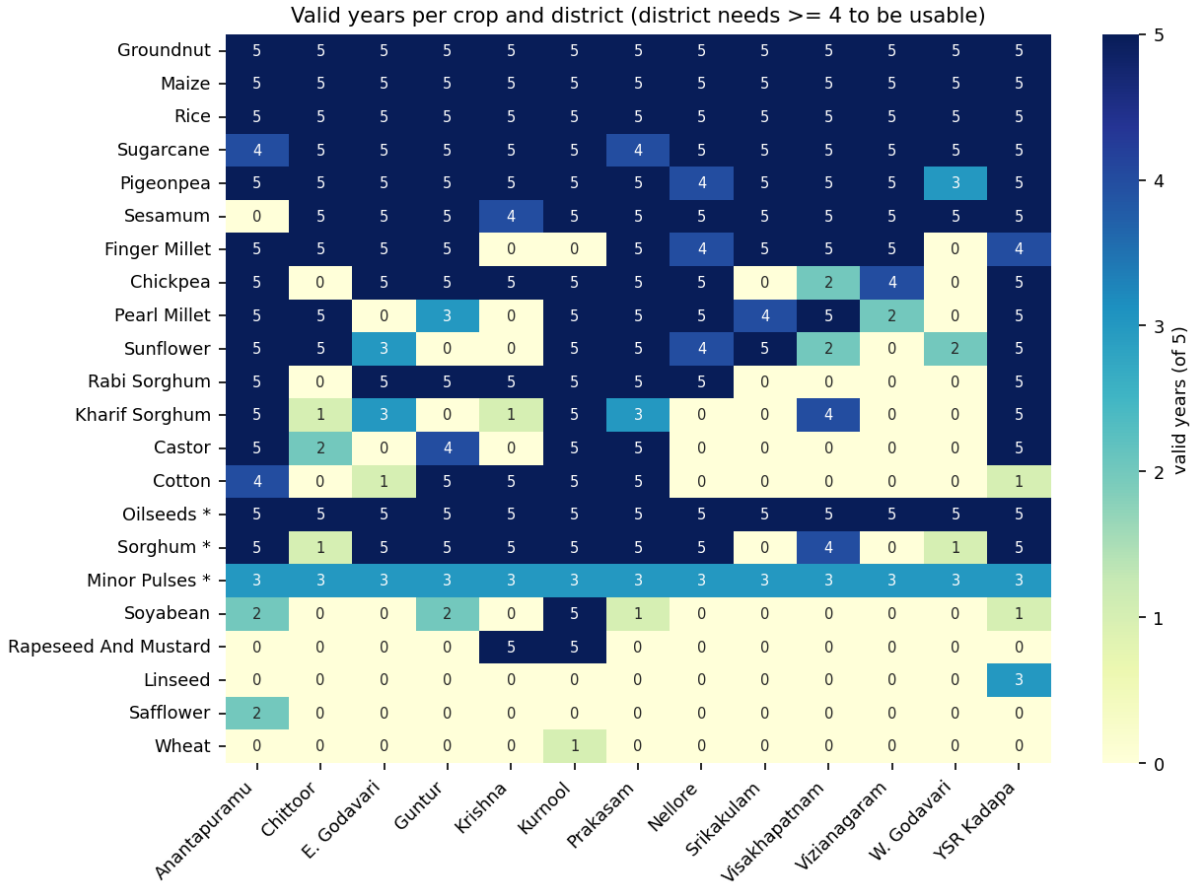


Figure 2. Valid years per crop and district.

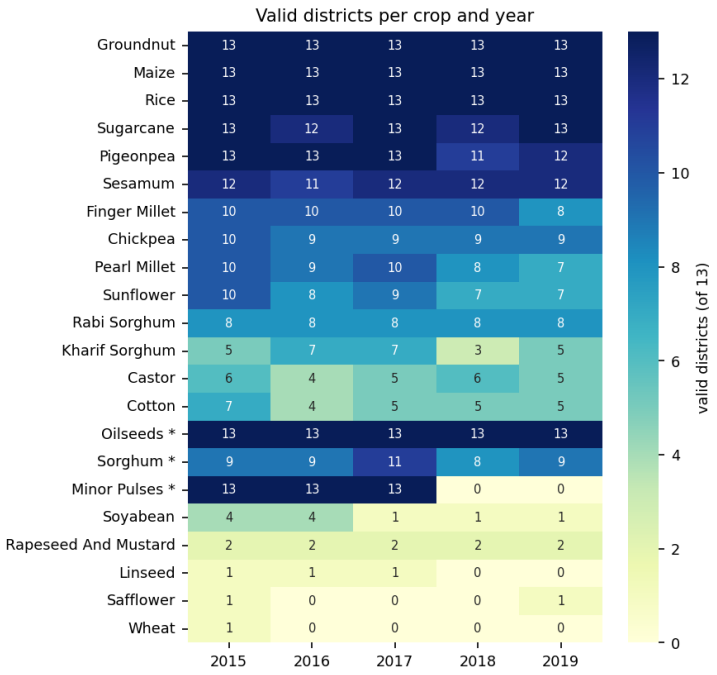


Figure 3. Valid districts per crop and year.

5. Minimum data requirements

Criterion	Eligible	Conditional	Ineligible	Rationale
Valid observations (N)	>= 30	15-29	< 15	Six candidate weather predictors (Tier 1 from Activity 4) at 5 observations per predictor; 3 predictors (water, heat, humidity) at 5 per predictor. Austin and Steyerberg (2015) found about 2 subjects per variable gives coefficient bias under 10% in linear regression; 5 is a 2.5x margin because Hanley (2016) notes 2 per variable is not enough when the model is used to predict
Usable districts (>= 4 valid years)	>= 6	3-5	< 3	Leave-one-year-out references need 3 training years; at least about half of the 13 districts so no single district dominates; 3 is the minimum to separate location from weather
Worst-year coverage	>= 5	3-4	< 3	Each held-out year needs enough districts for a stable error estimate
Area coverage	>= 80%	50-79%	< 50%	Valid observations must represent the crop (share of its total area)
Yield consistency	>= 90%	75-89%	< 75%	Rounding-aware agreement of yield with production / area among verifiable observations
Role (Issue 1)	include	-	aggregate / group category	Aggregates double count; minor pulses 2018-19 not reported

Classification rule: ineligible if any criterion is ineligible or the crop is excluded by role; conditionally eligible if none is ineligible but at least one is conditional; otherwise eligible. The thresholds are policy choices anchored in the rules above, not data-derived cut-offs, which is why the sensitivity analysis (Figure 7) is reported. Only **eligible** crops proceed; conditionally eligible crops need approval and the listed restrictions.

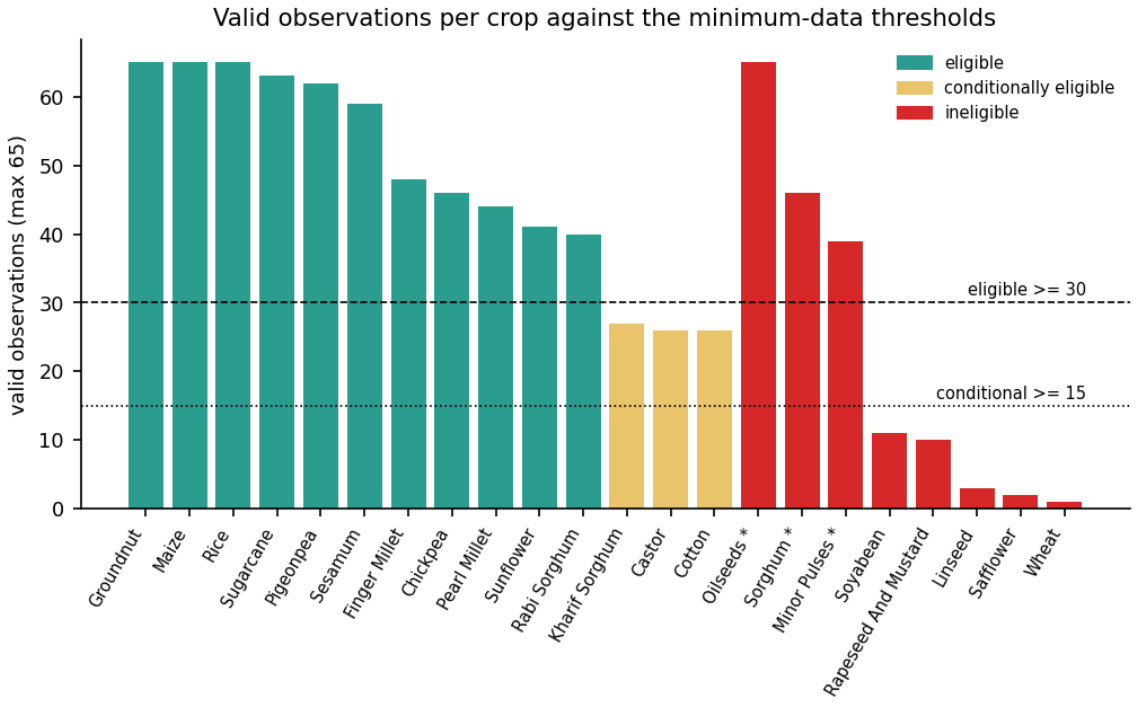


Figure 5. Valid observations per crop against the eligible and conditional thresholds.

6. Classification result

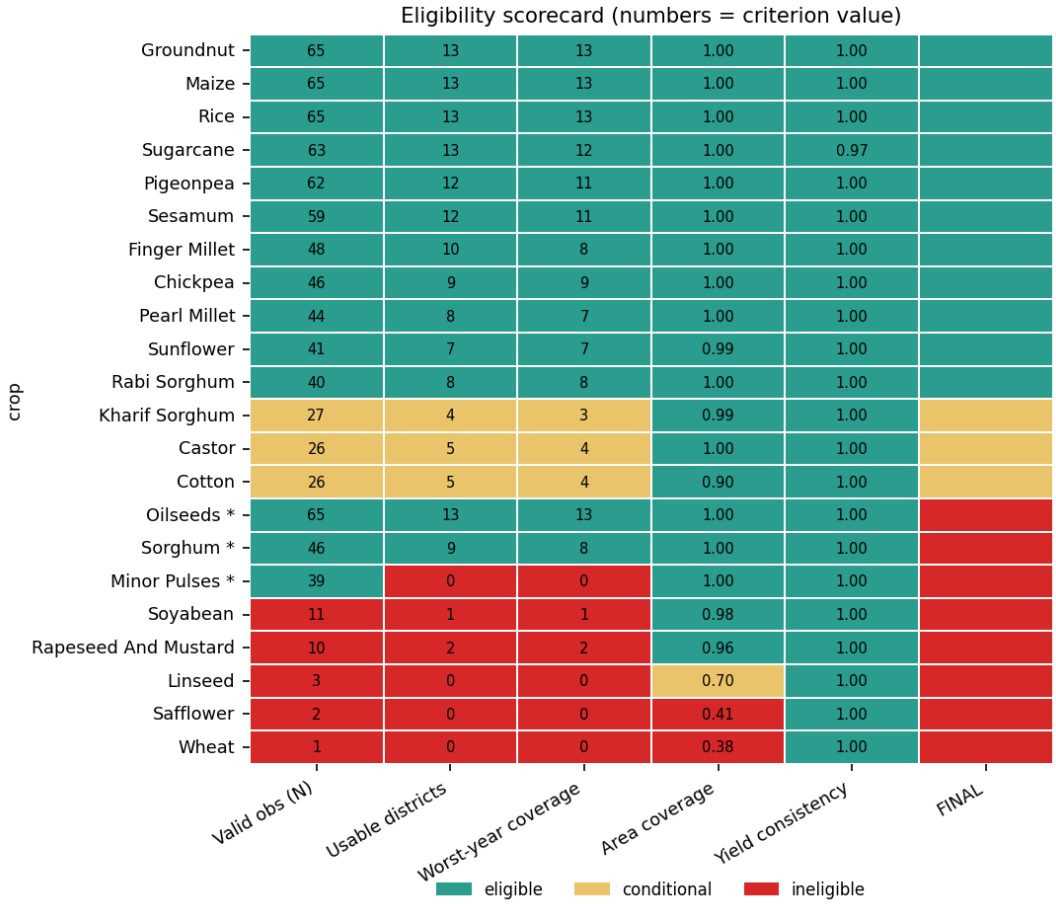


Figure 6. Eligibility scorecard: criterion values and bands per crop.

Crop	Class	Valid N	Usable districts	Worst year	Area cov.	Share of area	Reason / conditions
Groundnut	eligible	65	13	13	1.00	15.6%	meets all criteria
Maize	eligible	65	13	13	1.00	5.5%	meets all criteria
Rice	eligible	65	13	13	1.00	43.8%	meets all criteria
Sugarcane	eligible	63	13	12	1.00	2.0%	meets all criteria
Pigeonpea	eligible	62	12	11	1.00	5.3%	meets all criteria
Sesamum	eligible	59	12	11	1.00	1.0%	meets all criteria
Finger Millet	eligible	48	10	8	1.00	0.6%	meets all criteria
Chickpea	eligible	46	9	9	1.00	9.2%	meets all criteria
Pearl Millet	eligible	44	8	7	1.00	0.7%	meets all criteria
Sunflower	eligible	41	7	7	0.99	0.3%	meets all criteria
Rabi Sorghum	eligible	40	8	8	1.00	2.3%	meets all criteria
Kharif Sorghum	conditionally eligible	27	4	3	0.99	0.6%	borderline: N, districts, years. Conditions: limit to <= 5 predictors (5 obs per predictor); pooled model only; no district-specific reference; exclude years with < 5 valid districts from CV test folds; requires approval before modelling
Castor	conditionally eligible	26	5	4	1.00	0.8%	borderline: N, districts, years. Conditions: limit to <= 5 predictors (5 obs per predictor); pooled model only; no district-specific reference; exclude years with < 5 valid districts from CV test folds; requires approval before modelling
Cotton	conditionally eligible	26	5	4	0.90	12.2%	borderline: N, districts, years. Conditions: limit to <= 5 predictors (5 obs per predictor); pooled model only; no district-specific reference; exclude years with < 5 valid districts from CV test folds; requires approval before modelling

Crop	Class	Valid N	Usable districts	Worst year	Area cov.	Share of area	Reason / conditions
Oilseeds	ineligible	65	13	13	1.00	-	excluded by Issue 1 decision (exclude_aggregate)
Sorghum	ineligible	46	9	8	1.00	-	excluded by Issue 1 decision (exclude_aggregate)
Minor Pulses	ineligible	39	0	0	1.00	-	excluded by Issue 1 decision (exclude_group_category); fails: districts, years
Soyabean	ineligible	11	1	1	0.98	0.0%	fails: N, districts, years
Rapeseed And Mustard	ineligible	10	2	2	0.96	0.1%	fails: N, districts, years
Linseed	ineligible	3	0	0	0.70	0.0%	fails: N, districts, years; borderline: area_cov
Safflower	ineligible	2	0	0	0.41	0.0%	fails: N, districts, years, area_cov
Wheat	ineligible	1	0	0	0.38	0.0%	fails: N, districts, years, area_cov

- **Cotton** is conditional: only the five main districts (about 90% of cotton area) have verifiable, observed values, because the other districts report production in whole thousand tonnes and 2019 in eight districts is an allocated estimate. It is a major crop (12% of area), so supplying district-wise 2019 values would likely make it eligible.
- **Kharif sorghum and castor** are conditional on thin district coverage (4-5 usable districts).
- **Wheat, linseed, safflower, soyabean and rapeseed and mustard** have 1-11 valid observations in 0-2 usable districts and together make up under 0.2% of the modelling crop area; they are ineligible.

7. Sensitivity of the classification

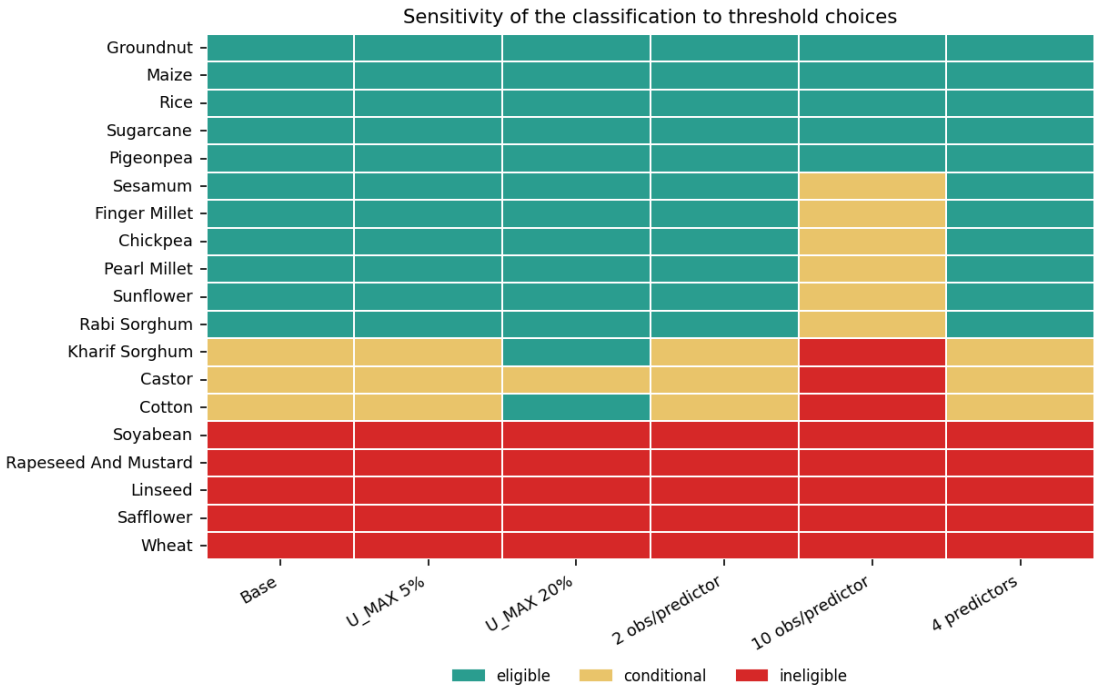


Figure 7. Class of each crop under alternative thresholds. Rice, maize, groundnut, sugarcane and pigeonpea never change.

8. Limitations

- Observations from the same district are correlated, so the effective sample size is smaller than the valid count; with 13 districts and 5 years the model evidence is thin even for eligible crops.
- The sample-size thresholds are rules of thumb from regression literature (Austin and Steyerberg 2015; Hanley 2016), not guarantees for this problem, and the minimum predictors per crop is an assumption (3 to 6).
- Valid-observation counts depend on the Issue 1 decisions (cotton 2019 estimates excluded, minor pulses excluded). Replacing the eight cotton 2019 estimates with official district values would change cotton's class.
- Materiality (share of area) is shown but not used as a criterion.

References

#	Source
[1]	Austin PC, Steyerberg EW (2015). The number of subjects per variable required in linear regression analyses. Journal of Clinical Epidemiology 68(6):627-636. doi:10.1016/j.jclinepi.2014.12.014
[2]	Hanley JA (2016). Simple and multiple linear regression: sample size considerations. Journal of Clinical Epidemiology 79:112-119. doi:10.1016/j.jclinepi.2016.05.014